

Krish Rajani

(587)-590-7817 | krajani@ualberta.ca | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

EDUCATION

University of Alberta

Expected Graduation: May 2027

BSc in Computing Science (Honors)

Edmonton, Canada

- Relevant Coursework: Algorithms, Software Engineering, Machine Learning, Computer Vision, File & Database Management, Object-Oriented Programming, Agile, DevOps, Networking

EXPERIENCE

Incoming Software Engineer Intern

May 2026 – Aug 2026

PCL Construction

Edmonton, Canada

- Developing enterprise data visualization tools in **C#** and **TypeScript** to support ML and data science initiatives

Software Development Intern

Sept 2025 – May 2026

Apex Utilities

Leduc, Canada

- Deployed a **RAG**-based chatbot using **Python**, **Azure**, and **PostgreSQL** vector storage, deflecting repetitive helpdesk tickets for 150 employees via a 0.8 cosine similarity threshold, reducing manual triage workload by **10%**
- Built a TF-IDF pipeline in **Python** to use **10K+** tickets to filter IT documentation and improve **RAG** quality by **70%**
- Indexed IT documentation using OpenAI embeddings via **PostgreSQL**, enabling semantic search on the knowledge base
- Containerized the **FastAPI** app with **Docker** and deployed to Azure, powering a Teams-integrated chat interface

Software Engineer Intern

May 2025 – Sept 2025

Wrise

Calgary, Canada

- Developed a platform in **React** and **TypeScript** to enable **500+** people to interact with creator-trained AI models
- Optimized video transcription pipeline by replacing Whisper API calls with SRT parsing and deduplication, cutting processing time by **60%** and reducing transcript token load by **70%**, thus lowering OpenAI API costs
- Built a **RAG** pipeline using **LangChain** and **GPT-4** to embed video transcripts into Supabase for contextual answers
- Integrated Stripe subscription flow with access gating, to re-enable chatbot access upon payment verification

PROJECTS

Core-Ledger-API | *Ruby on Rails, PostgreSQL, Minitest*

- Built a double-entry ledger API in **Ruby on Rails** and **PostgreSQL** with atomic transactions and pessimistic locking
- Designed idempotent REST endpoints with Bearer token auth, preventing duplicate transaction processing across retries
- Validated **30+** edge cases via **Minitest**, covering balance invariants, multi-currency conversion, and atomic rollback

TradePulse | *Java, Spring Boot, React, TypeScript*

- Built a full-stack trading strategy backtester using **Java** and **Spring Boot** with a **React + TypeScript** frontend, enabling users to simulate and visualize mean reversion strategies on historical market pricing data
- Deployed serverless backend using **AWS Lambda**, API Gateway, and **AWS DynamoDB** via **AWS SAM**
- Leveraged **JUnit** and **Mockito**, ensuring code reliability across market volatility and trade execution scenarios
- Designed RESTful APIs with JPA persistence, supporting backtest execution and historical result retrieval

FuelWise | *MongoDB (NoSQL), Express.js, Node.js, React.js*

- Engineered a fuel recommendation app using **React**, **Node.js**, and **MongoDB** that reduced fueling costs by **8%** for **100+** users, by implementing live station price analysis and geolocation-based optimization
- Orchestrated Google Maps API requests across Places, Details, and Distance Matrix endpoints with parallel batch processing, enabling real-time comparison of 40+ fuel stations per query in under **3 seconds**
- Architected a 3-tier caching system with request collapsing and adaptive rate limiting, cutting API costs by **90%**
- Accelerated release cycles by **20%** using GitHub Actions **CI/CD** for automated unit testing and deployment

Silent Signal (Hackathon) | *Python, React.js, AWS*

- Built a real-time sound classifier using **Python**, **React.js** and YAMNet to classify 520+ sounds with 95% accuracy
- Deployed serverless backend on **AWS Lambda**, API Gateway, and **DynamoDB** to log noise and automate alerts

LEADERSHIP EXPERIENCE

Assistant Director, Computing Science

May 2024 – May 2025

TeamUp Science

Edmonton, Canada

- Organized and taught Python to 20+ students during a workshop, improving coding fluency through live exercises
- Coordinated with mentors and volunteers to develop engaging lesson plans and emphasize collaboration between students

TECHNICAL SKILLS

Languages: Java, Python, Ruby, JavaScript, TypeScript, C, C#, C++, SQL

Frameworks & Libraries: Spring Boot, Ruby on Rails, FastAPI, React, Node, Express, HTML, CSS

Databases & Cloud: PostgreSQL, MySQL, MongoDB, AWS, Microsoft Azure, Google Cloud Platform

Developer Tools: Git, GitHub, Docker, Kubernetes, Redis, Postman, Linux